

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that **port 53 request is unreachable when attempting to visit the website.**

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: **“udp port 53 unreachable”**

The port noted in the error message is used for **DNS requests, which is a protocol that translates internet domain names into IP addresses. When a client computer wishes to access a website domain using their internet browser, a query is sent to the dedicated dns server. The dns server then look up for the ip that corresponds to that website domain.**

The most likely issue is **that there’s a problem with the web server (maybe it has been temporarily shut down) or the firewall or proxy server rules are stopping the access to port 53. Possibly even the long time of response to the requests can indicate some type of possible slow network traffic. It could be possible that a malicious attack happened on the website or that a possible threat is slowing our network.**

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident occurred **during the morning when several customers of clients reported that they were unable to access the client company website, this is how the IT department became aware of the incident.**

So the IT team responded and began running tests with the network protocol analyzer tool tcpdump. Results indicate that the incident is happening during the DNS network protocol as the “udp port 53 is unreachable”.

We are continuing monitoring the incident and analyzing the causes, in order to restore access again to the secure website.

The next steps include analyzing the firewall configuration and possible proxy server rules that may affect the accessibility of port 53. Then we are going to contact the system administrator of the web server in order to check the system if there’s signs of attacks and if the website is actually shut down. Controlling possible intrusions in

our network traffic it's going to be another step too, in order to verify if the possible slow traffic responses are caused by malicious actions inside our network.